This is RR

$$g_p^{rl}\sum_q\left(g_q^{ll}m^{ll}\overline{g_q^{ll}m^{rl}}+g_q^{ll}m^{lr}\overline{g_q^{rl}m^{rl}}+g_q^{lr}m^{ll}\overline{g_q^{lr}m^{rl}}+g_q^{lr}m^{lr}\overline{g_q^{rr}m^{rl}}+g_q^{rl}m^{ll}\overline{g_q^{ll}m^{rr}}+g_q^{rl}m^{lr}\overline{g_q^{rl}m^{rr}}+g_q^{rr}m^{ll}\overline{g_q^{lr}m^{rr}}+g_q^{rr}m^{lr}\overline{g_q^{rr}m^{rr}}\right)+$$
$$g_p^{rr}\sum_q\left(g_q^{ll}m^{rl}\overline{g_q^{ll}m^{rl}}+g_q^{ll}m^{rr}\overline{g_q^{rl}m^{rl}}+g_q^{lr}m^{rl}\overline{g_q^{lr}m^{rl}}+g_q^{lr}m^{rr}\overline{g_q^{rr}m^{rl}}+g_q^{rl}m^{rl}\overline{g_q^{ll}m^{rr}}+g_q^{rl}m^{rr}\overline{g_q^{rl}m^{rr}}+g_q^{rr}m^{rl}\overline{g_q^{lr}m^{rr}}+g_q^{rr}m^{rr}\overline{g_q^{rr}m^{rr}}\right)= \quad (1)$$
$$\sum_q d_{pq}^{rl}\left(g_q^{ll}\overline{m^{rl}}+g_q^{rl}\overline{m^{rr}}\right)+\sum_q d_{pq}^{rr}\left(g_q^{lr}\overline{m^{rl}}+g_q^{rr}\overline{m^{rr}}\right)$$

$$g_p^{rl}\sum_q\left(g_q^{ll}m^{ll}\overline{g_q^{ll}m^{ll}}+g_q^{ll}m^{lr}\overline{g_q^{rl}m^{ll}}+g_q^{lr}m^{ll}\overline{g_q^{lr}m^{ll}}+g_q^{lr}m^{lr}\overline{g_q^{rr}m^{ll}}+g_q^{rl}m^{ll}\overline{g_q^{ll}m^{lr}}+g_q^{rl}m^{lr}\overline{g_q^{rl}m^{lr}}+g_q^{rr}m^{ll}\overline{g_q^{lr}m^{lr}}+g_q^{rr}m^{lr}\overline{g_q^{rr}m^{lr}}\right)+$$
$$g_p^{rr}\sum_q\left(g_q^{ll}m^{rl}\overline{g_q^{ll}m^{ll}}+g_q^{ll}m^{rr}\overline{g_q^{rl}m^{ll}}+g_q^{lr}m^{rl}\overline{g_q^{lr}m^{ll}}+g_q^{lr}m^{rr}\overline{g_q^{rr}m^{ll}}+g_q^{rl}m^{rl}\overline{g_q^{ll}m^{lr}}+g_q^{rl}m^{rr}\overline{g_q^{rl}m^{lr}}+g_q^{rr}m^{rl}\overline{g_q^{lr}m^{lr}}+g_q^{rr}m^{rr}\overline{g_q^{rr}m^{lr}}\right)= \quad (2)$$
$$\sum_q d_{pq}^{rl}\left(g_q^{ll}\overline{m^{ll}}+g_q^{rl}\overline{m^{lr}}\right)+\sum_q d_{pq}^{rr}\left(g_q^{lr}\overline{m^{ll}}+g_q^{rr}\overline{m^{lr}}\right)$$

$$g_p^{ll}\sum_q\left(g_q^{ll}m^{ll}\overline{g_q^{ll}m^{rl}}+g_q^{ll}m^{lr}\overline{g_q^{rl}m^{rl}}+g_q^{lr}m^{ll}\overline{g_q^{lr}m^{rl}}+g_q^{lr}m^{lr}\overline{g_q^{rr}m^{rl}}+g_q^{rl}m^{ll}\overline{g_q^{ll}m^{rr}}+g_q^{rl}m^{lr}\overline{g_q^{rl}m^{rr}}+g_q^{rr}m^{ll}\overline{g_q^{lr}m^{rr}}+g_q^{rr}m^{lr}\overline{g_q^{rr}m^{rr}}\right)+$$
$$g_p^{lr}\sum_q\left(g_q^{ll}m^{rl}\overline{g_q^{ll}m^{rl}}+g_q^{ll}m^{rr}\overline{g_q^{rl}m^{rl}}+g_q^{lr}m^{rl}\overline{g_q^{lr}m^{rl}}+g_q^{lr}m^{rr}\overline{g_q^{rr}m^{rl}}+g_q^{rl}m^{rl}\overline{g_q^{ll}m^{rr}}+g_q^{rl}m^{rr}\overline{g_q^{rl}m^{rr}}+g_q^{rr}m^{rl}\overline{g_q^{lr}m^{rr}}+g_q^{rr}m^{rr}\overline{g_q^{rr}m^{rr}}\right)= \quad (3)$$
$$\sum_q d_{pq}^{ll}\left(g_q^{ll}\overline{m^{rl}}+g_q^{rl}\overline{m^{rr}}\right)+\sum_q d_{pq}^{lr}\left(g_q^{lr}\overline{m^{rl}}+g_q^{rr}\overline{m^{rr}}\right)$$

$$g_p^{ll}\sum_q\left(g_q^{ll}m^{ll}\overline{g_q^{ll}m^{ll}}+g_q^{ll}m^{lr}\overline{g_q^{rl}m^{ll}}+g_q^{lr}m^{ll}\overline{g_q^{lr}m^{ll}}+g_q^{lr}m^{lr}\overline{g_q^{rr}m^{ll}}+g_q^{rl}m^{ll}\overline{g_q^{ll}m^{lr}}+g_q^{rl}m^{lr}\overline{g_q^{rl}m^{lr}}+g_q^{rr}m^{ll}\overline{g_q^{lr}m^{lr}}+g_q^{rr}m^{lr}\overline{g_q^{rr}m^{lr}}\right)+$$
$$g_p^{lr}\sum_q\left(g_q^{ll}m^{rl}\overline{g_q^{ll}m^{ll}}+g_q^{ll}m^{rr}\overline{g_q^{rl}m^{ll}}+g_q^{lr}m^{rl}\overline{g_q^{lr}m^{ll}}+g_q^{lr}m^{rr}\overline{g_q^{rr}m^{ll}}+g_q^{rl}m^{rl}\overline{g_q^{ll}m^{lr}}+g_q^{rl}m^{rr}\overline{g_q^{rl}m^{lr}}+g_q^{rr}m^{rl}\overline{g_q^{lr}m^{lr}}+g_q^{rr}m^{rr}\overline{g_q^{rr}m^{lr}}\right)= \quad (4)$$
$$\sum_q d_{pq}^{ll}\left(g_q^{ll}\overline{m^{ll}}+g_q^{rl}\overline{m^{lr}}\right)+\sum_q d_{pq}^{lr}\left(g_q^{lr}\overline{m^{ll}}+g_q^{rr}\overline{m^{lr}}\right)$$